

Tutorial - 1

Topics: Preliminaries, Vectors

1. Let U be the universal set in a given context. Let P be a property for elements in U such that for each $x \in U$ either P holds, in which case we write, $P(x)$; or P doesn't hold in which case we write $\sim P(x)$. Let $P_T \subsetneq U$ be the subset for which P holds. That is, $P_T := \{x \in U : P(x)\}$ Prove the following :

(a) $[P \implies Q] \iff [P_T \subseteq Q_T]$

(b) $[P_T \subseteq Q_T] \iff [P_T^C \cup Q_T = U]$ and hence conclude,

$$[P \implies Q] \iff [\neg P \vee Q]$$

2. Suppose $\mathfrak{E}$ and $\mathbb{N}$ denotes the set of all even natural numbers and natural numbers respectively. Prove that if $p \in \mathbb{N}$ and $p^2 \in \mathfrak{E}$, then $p \in \mathfrak{E}$ using

(a) direct proof

(b) method of contradiction

(c) method of contraposition

3. Consider an initial investment of s in a T -period fixed deposit that pays $s(1+r)^{T-1}$ after T periods. This can be represented as a T -vector, viz., $b = [-s, 0, 0, \dots, 0, s(1+r)^{T-1}]$. Express b as a linear combination of $(T-1)$ one period unit loans where a one-period unit loan beginning in period $1 \leq t \leq T-1$ is denoted by $l^t = [0, \dots, 0, -1, (1+r), 0, \dots, 0]$. That is, find real numbers $\beta_1, \beta_2, \dots, \beta_{T-1}$ such that $b = \beta_1 l^1 + \beta_2 l^2 + \dots + \beta_{T-1} l^{T-1}$. Convince yourself that the answer corresponds to the idiom "To borrow from Peter to pay Paul".

4. Suppose $B = \{b^1, b^2, \dots, b^n\}$ form a basis of $\mathfrak{R}^n$. Further, suppose $0 \neq c = \alpha_1 b^1 + \alpha_2 b^2 + \dots + \alpha_n b^n$ is a non-zero vector in $\mathfrak{R}^n$. Prove that for any basis vector b^r for which $\alpha_r \neq 0$, is removed from the basis set B and in its place c is added, then $(B \setminus \{b^r\}) \cup \{c\}$ is a set of basis for $\mathfrak{R}^n$

5. Let $S := \{x^1, x^2, x^3\} \subset \mathfrak{R}^3$ be a set of linearly independent vectors. Let $S' = \{y^1, y^2, y^3\}$ be a new set of linearly independent set of vectors, generated from S as follows:

(a) $y^1 := \alpha_{11} x^1$

(b) $y^2 := \alpha_{21} x^1 + \alpha_{22} x^2$

(c) $y^3 := \alpha_{31} x^1 + \alpha_{32} x^2 + \alpha_{33} x^3$

Give a sufficient condition on α_{ij} where $i, j \in \{1, 2, 3\}$ for S' to be a set of linearly independent vectors

6. Let $x^1, x^2 \in \mathfrak{R}^d$ be non-zero vectors ($x^1, x^2 \neq 0$). Define the harmonic mean of their squared norms as:

$$H(x^1, x^2) := \frac{2\|x^1\|^2\|x^2\|^2}{\|x^1\|^2 + \|x^2\|^2}$$

Suppose x^1 and x^2 satisfy the dot-product condition:

$$0 \leq x^1 \cdot x^2 < H(x^1, x^2)$$

(a) **(AM–HM Bound)** Show that $H(x^1, x^2) \leq \frac{\|x^1\|^2 + \|x^2\|^2}{2}$, with equality if and only if $\|x^1\| = \|x^2\|$.

(b) **(Linear Independence)** Prove that $\{x^1, x^2\}$ is a linearly independent set.
